

Model Decoupling and Control of the Wheeled Humanoid Robot Moving in Sagittal Plane

Haitao Zhou, Xu Li, Haibo Feng, Jiachen Li, Songyuan Zhang, Yili Fu

Abstract—The wheeled humanoid robot is such a new type of robot that combines both the humanoid structure and the Wheeled Inverted Pendulum (WIP) base. They are able to move rapidly on flat ground as well as stand still on the slope, which has been well demonstrated on the WLR-II robot in this paper. In order to achieve it, a novel but simplified control framework is designed, which comprises of two main modules, the wheel balance controller and the centroidal adjustment controller. The former controller helps to maintain balance of the robotic system by rotating the wheel to move forward or backward, while the latter controller works by moving the Center of Mass (CoM) of the robot at a distance from the equilibrium point, which will result in a specified acceleration used to drive the first wheel balance controller. In order to design such these two controllers, the dynamic model of the robot in sagittal plane is decoupled into two relatively simplified model. In particular, the coupled dynamics between each other is significantly considered and alleviated. Experiments conducted on the WLR-II robot show that the proposed control framework can make the robot both accurately track the velocity trajectory and steadily stand on the slope.

I. INTRODUCTION

The wheeled humanoid robot appeared recently is a new type of humanoid robot as the driving wheel is in place of the plate foot structure usually adopted in the traditional humanoid or biped robot. Hence, the wheeled humanoid robots have to locomote by means of wheel rotation instead of stepping alternately. The Handle robot [1], released by the Boston Dynamics company in 2017, has attracted much attention in the filed of robotics. The released video has demonstrated its excellent mobility and maneuverability on horizontal ground, and even adaptability in complex terrain. However, the predecessors of wheeled humanoid robot can be tracked up to 2007. Engineers in Hitachi developed EMIEW-2 [2] service robot to apply it in indoor environment to interact with humans. Another wheeled humanoid robot, ASCENTO [3], is recently developed at ETH Zurich, with the goal of providing maximum indoor mobility. It is small in size and only weights 10 kg, but demonstrated prominent mobility performance. Inspired by the Handle robot, our laboratory also developed a hydraulic actuated wheeled humanoid robot WLR-I [4] in 2017, to explore mechanical

design, motion control and application in practice. WLR-II [5] designed in 2018 is another generation of the series of WLR robots. Unfortunately, most of these previous works are only concerning mechanical design, system integration as well as its ability demonstration and the problem of control and stability is rarely involved.

As for control algorithm, the wheeled humanoid robots are much closer related to the common two Wheeled Inverted Pendulum (WIP) robotic system [6-7] compared to the traditional humanoid robots, as both of these two types of robots have the wheel mechanics and move by continuous wheel rotation. In addition, the linear [6] or non-linear control controllers [7] for it are usually based on the precise dynamic dynamics and satisfying the Lyapunov function based stability requirements. However, as for control of the humanoid robots, it usually involves the problem of walking pattern generation based on a simplified Linear Inverted Pendulum (LIP) model [8] or foothold selection by the capture point [9]. As for the WIP robots, the controller design as well as its application to the real robotic system is very mature and great dynamic performance has achieved in various robots. The wheel balance control introduced in Section III. A is based on the equivalent WIP model in eq. (3), which is similar to the typical WIP model but subject to additional dynamic force and parameter variation from the decoupling process.

As the research scope in this paper is only limited in the control of the wheeled humanoid robot in sagittal plane, it can be regarded as a kind of wheeled manipulator [10-11]. The wheeled manipulator can make use of their redundant Degree of Freedom (DoF) above the wheel to change its CoM position to accelerate or decelerate the whole system. A related approach combining the classical feedback linearization with optimization method was proposed by Stilman et al. in [10] to control such an articulated mobile manipulator to accelerate, stand up and decelerate. It's the first time that the movement of internal mass strategy are applied to the robot to improve its dynamic performance. However, an explicit mass movement scheme is not given. Other wheel articulated mobile robots focused on implementing manipulation task or interacting with human. Kolhe et al. [11] presented three manipulation strategies for two wheel mobile manipulator to lift and push with heavy objects. In [12], an extended state observer (ESO) approach was adopted to estimate disturbance force and the estimated value was used to compensate it when implementing manipulation tasks.

The contributions presented in this paper mainly include two parts: Firstly, the dynamics of two wheeled humanoid

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robot is decoupled into two simplified models and the decoupled dynamics between these two models is well handled. Secondly, the wheeled balance controller and the centroidal adjustment controller, is separately established and successfully applied to the practical robotic system WLR-II.

This paper is organized as follows: modeling and decoupling of the wheeled humanoid robot is described in detail in Section II. Section III introduces the control framework for the wheeled humanoid robot. In Section IV, the control methods are conducted on the WLR-II robot and the results are discussed. Finally, concluding remarks are made in Section V.

TABLE I: SEVERAL PARAMETERS NOTATIONS

Symbol	Description
θ_w	Rotation angle of the driving wheel (in radians)
θ_b	Tilt angle of the body of the equivalent WIP model (in radians)
r	Radius of the driving wheel (in meters)
I_b	Total moments of inertia of above the driving wheel in the WIP model
l_c	Distance between the CoM position and wheel axis in the WIP model
c_w	Viscous damping coefficient of rotation of the wheel
$x_w, \dot{x}_w, \ddot{x}_w$	Displacement, velocity and acceleration of the driving wheel
$x_w^{ref}, \dot{x}_w^{ref}, \ddot{x}_w^{ref}$	Desired displacement, velocity and acceleration of the driving wheel

II. MODEL PREPARATION

A. Modeling and Decoupling

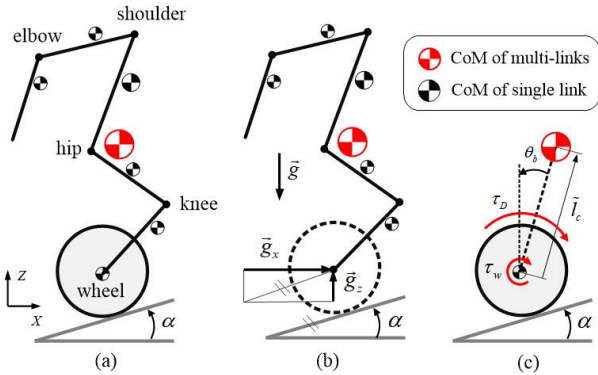


Fig. 1: Schematics of three different types of dynamic model of robot when it stands on the slope. (a) Wheeled humanoid robot in sagittal plane; (b) The decoupled multi-body model with a floating base; (c) The decoupled equivalent WIP model.

Define the generalized vector for the robotic system WLR-II as $\mathbf{q}$. In order to decouple the whole dynamics, the generalized vector $\mathbf{q} = [\mathbf{q}_w; \mathbf{q}_h]^T$ is divided into two state vectors $\mathbf{q}_w$ and $\mathbf{q}_h$, which are established for the wheel segment and the multi-body segment, respectively. The dynamics for the WLR-II in sagittal plane on the slope, as shown in Fig. 1

(a), can be represented by

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{G}(\mathbf{q}) = \mathbf{J}^T(\mathbf{q})\boldsymbol{\tau} \quad (1)$$

where $\mathbf{M}(\mathbf{q}) \in \mathbb{R}^{(n+3) \times (n+3)}$ is the symmetric, positive definite inertia matrix, $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \in \mathbb{R}^{n+3}$ is the vector related to the Coriolis and centrifugal forces and $\mathbf{G}(\mathbf{q}) \in \mathbb{R}^{n+3}$ is the vector of gravitational forces, $\mathbf{J}^T(\mathbf{q}) \in \mathbb{R}^{(n+3) \times n}$ is the matrix of control coefficients and $\boldsymbol{\tau} = [\tau_w, \tau_2, \tau_3, \dots, \tau_n]^T \in \mathbb{R}^n$ is the n -dimensional vector of the control inputs. According to dynamic property, each term in eq. (1) might incorporate two segments, the wheel part and the multi-body part, given by

$$\mathbf{M}(\mathbf{q}) = \begin{bmatrix} \mathbf{M}_{ww} & \mathbf{M}_{wh} \\ \mathbf{M}_{hw} & \mathbf{M}_{hh} \end{bmatrix}; \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) = \begin{bmatrix} \mathbf{C}_w \\ \mathbf{C}_h \end{bmatrix}$$

$$\mathbf{G}(\mathbf{q}) = \begin{bmatrix} \mathbf{G}_w \\ \mathbf{G}_h \end{bmatrix}; \boldsymbol{\tau} = [\tau_w \quad \tau_h]$$
(2)

Model decoupling is a valid method to simplify controller design process for complex dynamic system but can retain the same excellent quality [19]. On the basis of eq. (1) and eq. (2), the whole dynamics will be decoupled into two segments in the following part, with respect to the equivalent WIP model and the multi-body model with a floating base, respectively.

B. The Equivalent WIP Model

The equivalent WIP model, as shown in Fig. 1 (c), is established by regarding the multi-body segments as a single intermediate body of the typical WIP model. However, some difference exists between them, as the equivalent WIP model may subject to the parameter variation and dynamic force arising from the movement of multi-body segment. The variable parameter set can be denoted as a vector defined as $\boldsymbol{\vartheta} = [\tilde{l}_b, \tilde{l}_c]^T$, containing two elements, the position of CoM $\tilde{l}_c$ and the moment of inertia $\tilde{l}_b$, which is a function of joints angle of the multi-body segments and will be selected as a gain scheduling vector in Section IV. On the other hand, the dynamic force τ_D as shown both in Fig. 1 (c) and in eq. (3) arising from the joint velocity and acceleration of the multi-body segments.

Similar to the traditional WIP model case, select $\mathbf{q}_w = [\theta_w, \theta_b]^T$ as the generalized state vector and the dynamics of the equivalent WIP model on the slope can be written in the form

$$\mathbf{M}_w(\mathbf{q})\ddot{\mathbf{q}}_w + \mathbf{C}_w(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{G}_w(\mathbf{q}) = \mathbf{J}_w^T \boldsymbol{\tau}_w + \boldsymbol{\tau}_D \quad (3)$$

where $\mathbf{M}_w(\mathbf{q}) \in \mathbb{R}^{2 \times 2}$, $\mathbf{C}_w(\mathbf{q}, \dot{\mathbf{q}}) \in \mathbb{R}^{2 \times 1}$, $\mathbf{G}_w(\mathbf{q}) \in \mathbb{R}^{2 \times 1}$, $\mathbf{J}_w(\mathbf{q}) \in \mathbb{R}^{2 \times 1}$ and $\boldsymbol{\tau}_D \in \mathbb{R}^{2 \times 1}$ given by the expressions

$$\mathbf{M}_w(\mathbf{q}) = \begin{bmatrix} \tilde{l}_b + m_b \tilde{l}_c^2 & m_b \tilde{l}_c \cos(\theta_b + \alpha) \\ m_b \tilde{l}_c \cos(\theta_b + \alpha) & I_w + m_b r^2 + m_w r^2 \end{bmatrix}$$

$$\mathbf{C}_w(\mathbf{q}, \dot{\mathbf{q}}) = \begin{bmatrix} -c_w \dot{\theta}_w + c_w \dot{\theta}_b \\ -m_b \tilde{l}_c \sin \theta_b \cdot \dot{\theta}_b^2 + c_w \dot{\theta}_w - c_w \dot{\theta}_b \end{bmatrix}$$

$$\mathbf{G}_w(\mathbf{q}) = \begin{bmatrix} -m_b g \tilde{l}_c \sin \theta_b \\ 0 \end{bmatrix}, \mathbf{J}_w(\mathbf{q}) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \boldsymbol{\tau}_D = \begin{bmatrix} \tau_{D1} \\ \tau_{D2} \end{bmatrix}$$

The dynamic force, also referred to as the disturbance force to the equivalent WIP model, can be represented as a nonlinear function about the joint velocity and acceleration in the following form

$$\begin{aligned}\tau_{D1} &= \Gamma_1(\mathbf{q}_h, \dot{\mathbf{q}}_h, \ddot{\mathbf{q}}_h), \tau_{D2} = \Gamma_2(\mathbf{q}_h, \dot{\mathbf{q}}_h, \ddot{\mathbf{q}}_h) \\ \mathbf{q}_h &\in [\underline{\mathbf{q}}_h, \bar{\mathbf{q}}_h], \dot{\mathbf{q}}_h \in [\underline{\dot{\mathbf{q}}}_h, \bar{\dot{\mathbf{q}}}_h], \ddot{\mathbf{q}}_h \in [\underline{\ddot{\mathbf{q}}}_h, \bar{\ddot{\mathbf{q}}}_h]\end{aligned}\quad (4)$$

From eq. (4), we can derive that the disturbance force is limited by the upper and lower bound as $\tau_D \in [\tau_{D,\min}, \tau_{D,\max}]$.

C. The Multi-body Model with a Floating Base

In order to design controller for the multi-body segments in joint space $\mathbf{q}_h$, the multi-body model with a floating base is created, which consists of a series of rigid links with the total number of n . Compared to the typical multi-link model with the fixed base, the floating base type studied is governed by several underactuated degree of freedoms dependent on the movement constraints. As shown in Fig. 1 (b), this model will suffer from an additional acceleration $\ddot{g}_x$ and $\ddot{g}_z$ at the wheel centre along the direction of x and z , respectively.

Select the generalized state vector as $\mathbf{q}_h = [\theta_1, \theta_2, \dots, \theta_n]^T$. Based on the Euler-Lagrange method, the dynamical equation about the decoupled multi-body model with a floating base is derived as

$$\begin{aligned}\mathbf{M}_h(\mathbf{q}_h)\ddot{\mathbf{q}}_h + \mathbf{C}_h(\mathbf{q}_h, \dot{\mathbf{q}}_h) + \mathbf{G}_h^g(\mathbf{q}_h)|\ddot{g}| + \mathbf{G}_h^x(\mathbf{q}_h)|\ddot{g}_x| \\ + \mathbf{G}_h^z(\mathbf{q}_h)|\ddot{g}_z| = \mathbf{J}_h^T \cdot \tau_h\end{aligned}\quad (5)$$

where $\mathbf{M}_h(\mathbf{q}_h) \in \mathbb{R}^{n \times n}$ are the inertia force terms; $\mathbf{C}_h(\mathbf{q}_h, \dot{\mathbf{q}}_h) \in \mathbb{R}^n$ are the Coriolis and centrifugal forces terms; $\mathbf{G}_h^g(\mathbf{q}_h)|\ddot{g}| \in \mathbb{R}^n$ are the gravitational forces terms. $\mathbf{J}_h^T \in \mathbb{R}^{n \times (n-1)}$ is the select matrix of control inputs. $\tau_h = [\tau_2, \tau_3, \dots, \tau_n] \in \mathbb{R}^{1 \times (n-1)}$ is the control inputs. $\mathbf{G}_h^x(\mathbf{q}_h)|\ddot{g}_x| \in \mathbb{R}^n$ and $\mathbf{G}_h^z(\mathbf{q}_h)|\ddot{g}_z| \in \mathbb{R}^n$ are the additional acceleration terms induced by the movement of floating base, the resultant of these two terms point in the direction perpendicular to the slope surface. The additional acceleration term is bounded by

$$\begin{aligned}\underline{G}|\ddot{g}_x|_{\min} \leq \mathbf{G}_h^x(\mathbf{q}_h)|\ddot{g}_x| \leq \bar{G}|\ddot{g}_x|_{\max} \\ \underline{G}|\ddot{g}_z|_{\min} \leq \mathbf{G}_h^z(\mathbf{q}_h)|\ddot{g}_z| \leq \bar{G}|\ddot{g}_z|_{\max}\end{aligned}\quad (6)$$

where the $\mathbf{G}_h^x(\mathbf{q}_h)$ term is bounded by $\underline{G}$ and $\bar{G}$ determined by the joint angle vector $\mathbf{q}_h$ of the multi-links.

D. Problem Statement

The control objectives for the wheeled humanoid robots can be stated in the following two aspects:

1) Asymptotic stabilization of the reference movement trajectory of velocity or position $\dot{\theta}_w^{ref}(t)$, $\theta_w^{ref}(t)$ in Cartesian space, satisfying

$$\lim_{t \rightarrow +\infty} |\dot{\theta}_w(t) - \dot{\theta}_w^{ref}(t)| = 0, \lim_{t \rightarrow +\infty} |\theta_w(t) - \theta_w^{ref}(t)| = 0 \quad (7)$$

2) Keep small, bounded errors between the desired and the real joint trajectory in joint space, defined as $|\mathbf{q}_h(t) - \mathbf{q}_h^{ref}(t)| \leq \epsilon$, where ϵ is an arbitrary small positive constant.

III. PROPOSED CONTROL METHOD

A. Wheel Balance Control

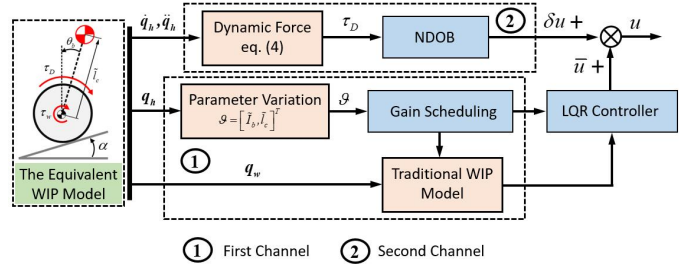


Fig. 2: The architecture of the wheel balance control.

On the basis of the decoupled equivalent WIP model in Section II, the wheel balance controller can then be designed as illustrated in Fig. 2. The decoupled equivalent WIP model include three parts: the traditional WIP model, the dynamic force (disturbance force) and the parameter variation. And two channels are established to handle it and calculate the control input. In the first channel, the linear quadratic regulator (LQR) method is used to design the baseline controller to guarantee the optimal performance in current operating point, while the gain scheduling approach is adopted to update the WIP model by these variable parameters $\tilde{l}_b$ and $\tilde{l}_c$ (also referred to schedulable variables in control theory). In the second channel, another disturbance, dynamic force, is observed by nonlinear disturbance observer method (NDOB) and is then compensated by the additional torque input δu . Therefore, the total control input, also referred to the desired wheel torque τ_w , is calculated as $u = \bar{u} + \delta u$.

LQR based gain scheduling method

Choose the state variable as $\mathbf{x} = [x_1, x_2, x_3, x_4]^T = [\theta_b, \dot{\theta}_b, \theta_w, \dot{\theta}_w]^T$. Select the gain scheduling vector as the variable parameters $\vartheta = [\tilde{l}_b, \tilde{l}_c]^T$. The dynamics of the equivalent WIP model in eq. (3) can be rewritten as

$$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}, \vartheta) + \mathbf{G}(\mathbf{x}, \vartheta)u + \mathbf{D}(\mathbf{x}, \vartheta) \quad (8)$$

where $\mathbf{F}(\mathbf{x}, \vartheta) = [x_2, \mathbf{f}_b(\mathbf{x}, \vartheta), x_4, \mathbf{f}_w(\mathbf{x}, \vartheta)]$ and $\mathbf{G}(\mathbf{x}, \vartheta) = [0, \mathbf{g}_b(\mathbf{x}, \vartheta), 0, \mathbf{g}_w(\mathbf{x}, \vartheta)]$ are the nonlinear function vectors. The nonlinear disturbance function is $\mathbf{D}(\mathbf{x}, \vartheta) = [0, \mathbf{d}_b(\mathbf{x}, \vartheta), 0, \mathbf{d}_w(\mathbf{x}, \vartheta)]$.

Linearize eq. (8) at the equilibrium point by means of Taylor expansion method with the first terms retained, then we obtain

$$\begin{aligned}\dot{\mathbf{x}} &= \frac{\partial \mathbf{F}(\mathbf{x}, \vartheta)}{\partial \mathbf{x}} \bigg|_{\mathbf{x}=\mathbf{x}^*} (\mathbf{x} - \mathbf{x}^*) + \\ &\quad \frac{\partial \mathbf{G}(\mathbf{x}, \vartheta)}{\partial \mathbf{x}} \bigg|_{\mathbf{x}=\mathbf{x}^*} (\mathbf{x} - \mathbf{x}^*) u + \mathbf{D}(\mathbf{x}, \vartheta)\end{aligned}\quad (9)$$

The linearized state space equation of WIP system can be derived as

$$\dot{\mathbf{x}} = \mathbf{A}(\vartheta)\mathbf{x} + \mathbf{B}(\vartheta)u + \mathbf{D}(\mathbf{x}, \vartheta) \quad (10)$$

where $\mathbf{A}(\vartheta)$, $\mathbf{B}(\vartheta)$ are the system matrix and output vector. Both of these are the nonlinear function with respect to the gain scheduling vector ϑ .

LQR approach is widely used to design the controller for the WIP model, as it is easy to compute and have excellent dynamic performance. The objective of this control method is to minimize the performance index

$$J = \int_0^\infty [\tilde{\mathbf{x}}^T \mathbf{Q} \tilde{\mathbf{x}} + u^T R u] dt \quad (11)$$

Where the state error vector is represented by $\tilde{\mathbf{x}} = \mathbf{x} - \mathbf{x}_d$; $\mathbf{Q}$ and R are the weigh matrices that determining the performance of the LQR controller.

$$\bar{u} = -\mathbf{K} \cdot \tilde{\mathbf{x}} \quad (12)$$

where the feedback gain vector is $\mathbf{K} = [k_1, k_2, k_3, k_4]^T$ and $\bar{u}$ is the control input computed from the baseline controller without consideration of disturbance. Note that the gain vector $\mathbf{K}$ is calculated online according to the variation of the gain scheduling vector.

NDOB Method

The nonlinear disturbance observer (NDOB) [15] [16] is explored to estimate the slowly time varying disturbance, which is expressed as

$$\begin{cases} \dot{\mathbf{z}} = -\mathbf{l}(\mathbf{x})\mathbf{z} - \mathbf{l}(\mathbf{x})[\mathbf{p}(\mathbf{x}) + \tilde{\mathbf{F}}(\mathbf{x}, \vartheta) + \tilde{\mathbf{G}}(\mathbf{x}, \vartheta)u] \\ \hat{\mathbf{d}} = \mathbf{z} + \mathbf{p}(\mathbf{x}) \end{cases} \quad (13)$$

where $\mathbf{z} = [z_1, z_2]^T$ is the internal state of the nonlinear disturbance observer; $\mathbf{l}(\mathbf{x}) = [l_1(x), l_2(x)]^T$ is the observer gain vector to be designed; $\hat{\mathbf{d}}$ is the estimated disturbance force; $\tilde{\mathbf{F}}(\mathbf{x}, \vartheta) = [\mathbf{f}_b(\mathbf{x}, \vartheta), \mathbf{f}_w(\mathbf{x}, \vartheta)]^T$ and $\tilde{\mathbf{G}}(\mathbf{x}, \vartheta) = [\mathbf{g}_b(\mathbf{x}, \vartheta), \mathbf{g}_w(\mathbf{x}, \vartheta)]^T$ are the dynamic system term as indicated in eq. (8). The NDOB gain $l(x)$ is determined by

$$\mathbf{l}(\mathbf{x}) = \frac{\partial \mathbf{p}(\mathbf{x})}{\partial \mathbf{x}} \quad (14)$$

We choose $\mathbf{p}(\mathbf{x}) = [\theta_b, \theta_w]^T$, which can guarantee the asymptotical convergence of the estimated disturbance. The compensation input torque of the driving wheel can be expressed as

$$\delta u = \mathbf{K}_{\text{ndob}} \cdot \hat{\mathbf{D}} \quad (15)$$

where $\mathbf{K}_{\text{ndob}} = [k_{\text{ndob}1}, k_{\text{ndob}2}]$ is the feedback gain with respect to the inverse of the inertia term. $\hat{\mathbf{D}}$ is the estimated disturbance applied to the system.

B. Centroidal Adjustment Control

Effect of the measurement error of CoM position on the WIP system

The traditional feedback control law for the WIP model can be written as the product of the feedback gain vector and the state error vector as shown in eq. (12). To be further, we expand the form of multiplication of vectors as follows

$$\tau_w = \underbrace{k_1(\theta_b - \theta_b^{ref})}_{\tau_{w, \theta_b}} + \underbrace{k_2(\dot{\theta}_b - \dot{\theta}_b^{ref})}_{\tau_{w, \dot{\theta}_b}} + \underbrace{k_3(\dot{\theta}_w - \dot{\theta}_w^{ref})}_{\tau_{w, \dot{\theta}_w}} \quad (16)$$

where τ_w is the expected wheel motor torque. As we only consider the case of velocity tracking control of the WIP

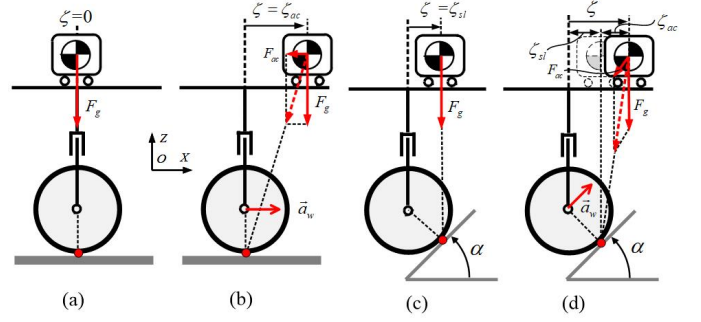


Fig. 3: The centroidal behavior of the wheeled inverted pendulum model with a sliding mass (referred to the CoM of the body) in four different movement cases. (a) There is no displacement of CoM for this model on the horizontal ground without acceleration; (b) The displacement of CoM is used to generate an acceleration of the system; (c) The displacement of CoM is used to compensate for the slope angle; (d) The displacement of CoM is used to both compensate for the slope angle and generate a movement acceleration.

model, the wheel motor torque totally incorporate three terms induced by the tilt angle error, angular velocity error and velocity error, respectively, given as

$$\tau_w = \tau_{w, \theta_b} + \tau_{w, \dot{\theta}_b} + \tau_{w, \dot{\theta}_w} \quad (17)$$

Assume that there is a deviation between the real CoM of the body and the measured value. In reality, this error resulting from its measurement mainly change the desired tilt angle θ_b^{ref} (computed by the measurement) instead of the CoM distance l_c . So the error of the tilt angle reference $\delta\theta_b^{ref}$ applied to the nominal system will be analyzed and it can be denoted as

$$\theta_b^{ref} = \theta_b^{ref*} + \delta\theta_b^{ref} \quad (18)$$

Since the real reference tilt angle θ_b^{ref*} is always selected as the posture to keep the CoM above the wheel axis in the vertical direction, the existing error $\delta\theta_b^{ref}$ will cause an error to the torque term τ_{w, θ_b} . Suppose that this model finally move in a constant velocity on horizontal ground, $\tau_w = 0$ is satisfied. As τ_{w, θ_b} is not zero, the third term in eq. (17) $\tau_{w, \dot{\theta}_w}$ must be exploited to keep $\tau_w = 0$. Combining eq. (18) and eq. (16) as well as $\tau_w = 0$, we derived the following equation as

$$-k_1\delta\theta_b^{ref} + k_3\delta\dot{\theta}_w = 0 \quad (19)$$

Where we conclude that the reference tilt angle error is proportional to the error of real tilt angle.

CoM compensation + PD control

From eq. (19), we conclude that the active displacement of CoM can facilitate to change the movement velocity. In reality, the displacement of CoM is directly related to the acceleration of the system rather than the velocity. This principle will be used in control of the wheeled humanoid robot. For simplicity, the complete model is simplified as a WIP model with a sliding mass [13,14], as shown in Fig.

3, where the sliding mass (CoM) is only allowed to move along the horizontal direction relative to wheel axis, while the driving wheel is assumed massless.

In Fig. 3, we consider two cases that the robot move on horizontal ground and the slope, respectively. The stability principle of the Inverted Pendulum (IP) model in [17] is introduced, which states that the resultant force acting on the CoM, including the gravitational force and the inertial force, must point toward and go through the support polygon. Then, in this case the IP type model will not suffer from an extra moment with respect to the contact point and will not topple over. This principle is completely suitable for the WIP model. The CoM control law can be directly given as

$$\zeta = \zeta_{sl} + \zeta_{ac} + \Delta\zeta \quad (20)$$

where ζ_{sl} is the CoM displacement used to compensate for the slop angle; ζ_{ac} is the CoM displacement to generate an specified acceleration; $\Delta\zeta$ is a PD feedback control law used to reduce the feedforward compensate error. To be further, the detail expressions about these terms are given as

$$\begin{aligned} \zeta_{sl} &= r \sin \alpha \\ \zeta_{ac} &= K_{com} \ddot{\theta}_w^{ref} \\ \Delta\zeta &= K_p^\zeta (\theta_w - \theta_w^{ref}) + K_d^\zeta (\dot{\theta}_w - \dot{\theta}_w^{ref}) \end{aligned} \quad (21)$$

where r is the wheel radius and α is the slope angle. K_{com} is the feedforward gain related to the wheel angular velocity. K_p^ζ and K_d^ζ are standard PD feedback gains with respect to CoM control law. They are adjusted according to the response result in our experiment.

IV. EXPERIMENTAL RESULTS

A. WLR-II Robot

Some experiments are conducted on the wheel-legged robot WLR-II [5], as shown in Fig. 4. The robot is 180 cm in height with a total mass of 70 kg. The robot has 13 DoFs (Degree of Freedom), where 3 DoFs in each leg, 1 DoF in the waist and 3 DoFs in each arm. Except for the driving wheels, all the joints are driven by hydraulic actuators to provide larger torque and excellent dynamic response. An IMU sensor (Inertia Measurement Unit) is installed on the bottom of the waist to measure the body posture. The dynamic parameters of body parts of the real robot is given in Table II. To verify the control framework introduced above, We conducted experiments in two different cases to demonstrate the proposed control algorithm. In scenario I, the robot is controlled to track the desired acceleration-continuous velocity profiles on horizontal ground. In scenario II, the robot is controlled to move to the slope with unexpected inclination and stand still on it. In addition, we set the CoM position in vertical direction with respect to the driving wheel as $0.55m$, which is the normal working case of the robot.

B. Velocity Tracking on Horizontal Ground

In order to ensure smoothness, the polynomial splines is used to generate the desired velocity trajectory in advance. This whole motion process can be divided into three phases:

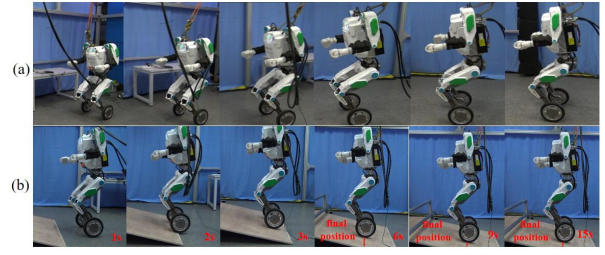


Fig. 4: The experiment snapshots of the WLR-II robot. (a) Locomotion on horizontal ground following a planned velocity profiles. (b) Moving up a wooden ramp and stopping at a designated location.

TABLE II: DYNAMIC PARAMETERS OF WLR-II

Body part	Mass [kg]	Length[m]	Moment of inertia [kg·m ²]	CoM Position [m]
Driving wheel	$m_w = 3.2$	$r = 0.12$	$I_w = 0.0474$	*
Shank	$m_1 = 11.0$	$l_1 = 0.40$	$I_1 = 0.1921$	$l_{c1} = 0.15$
Thigh	$m_2 = 13.2$	$l_2 = 0.40$	$I_2 = 0.2426$	$l_{c2} = 0.24$
Upper body	$m_3 = 42.6$	*	$I_3 = 2.5113$	$l_{c3} = 0.35$

acceleration phase, constant speed phase and deceleration phase. The desired trajectory of movement is represented by Fig. 5 (a). The maximal acceleration and velocity are separately selected as $1m/s^2$ and $0.6m/s$, which is assumed to be the normal working condition for the wheel-legged robot to implement tasks. It's found that the robot was unable to track the reference velocity profiles well without centroidal adjustment, especially when the required acceleration is higher than a certain range. However, with centroidal adjustment the robot achieved a high velocity tracking accuracy during the whole process of movement. The maximum speed tracking error of robot is no more than $0.05m/s^2$.

In Fig. 5 (b), the changes of joint angle during a motion cycle is described, which indicates the centroidal adjustment is working to keep balance. The knee joint and hip joint are approximately 85° and 45° , respectively. Both of these two

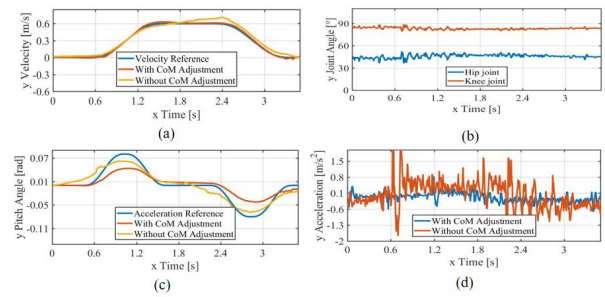


Fig. 5: The experimental results of robot moving on horizontal ground during a motion cycle. (a) Velocity tracking results. (b) The measurements of knee and hip joint angle. (c) The measurements of pitch angle. (d) The measurements of forward acceleration.

joint angle vary in a small range indicating there is no large disturbance .

As shown in Fig. 5 (c) and Fig. 5 (d), the pitch angle errors and the forward acceleration in a motion cycle were described by comparison of these two cases, which indicates that the centroidal adjustment scheme can decrease the body shaking of the robot and improve the stability of the robot during the movement.

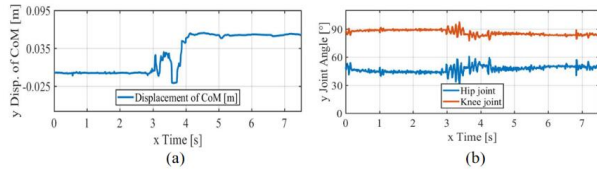


Fig. 6: The experimental results of robot climbing up a slope. (a) The measurement of displacement of CoM. (b) The measurement of knee and hip joint angle.

C. Standing Still on the Slope

A slope consisting of a inclined board and aluminum alloy frame serves as the experimental platform. Since there is no vision sensor installed on the robot, the information of slope is unknown to it. In the experiment, the robot stands still on horizontal ground in a distance from the slope, then it move forward according to the given demands until arrived at the designated location on the slope. The tilt angle of the slope measured is about 10° . The robot can maintains its desired position and equilibrium state only relying on the control algorithm proposed in this paper.

It's found in Fig. 6 (a) that the transition from flat ground to the slope happens about in the third second for the robot. On horizontal ground, the CoM position of the robot in X direction is about $-0.025m$, while the value of it becomes approximately $0.055m$. Hence, there is a fixed deviation about $0.08m$ between these two types of ground with different inclination angles.

As shown in Fig. 6 (b), when the robot moved on the slope, there is a constant change in the knee and hip joint angle. On the slope, the robot moved its CoM to the desired equilibrium point to keep balance. Note that in our experiment the robot was unable to stand steadily still on the slope without the centroidal adjustment, even though we have attempted to increase the feedback gains of wheeled balance controller.

V. CONCLUSIONS

This paper has proposed a valid control framework for a novel wheeled humanoid robot, which consists of a wheeled balance controller and a centroidal adjustment controller. The wheeled balance controller is used to enable the robot to maintain balance by rotate its wheel, while the centroidal adjustment controller moves the CoM of the robot to improve its control accuracy and stability. Eventhough only the wheeled balance controller can also make the robot keep its equilibrium in horizontal ground, it has limited control accuracy in the presence of disturbance and can't implement

complicated task, such as move on the slope. So we are sure that the centroidal adjustment controller plays an significant role in control of two wheel-legged robot to further improve its performance.

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